
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Collaboration Systems
Manufacturer: Bosch
Firmware Version: N/A
Model(s): Dicentis DCNM-LSYS

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.09.0000-b009	2.5.0

Version History

Module Version	Date	Notes
1_2_2_0	7/22/2020	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`
- NumberofActiveMicrophones variable must be set accordingly. Default value is '5'.
NumberofActiveMicrophones ranges from 1 to 10.
Example: `InterfaceName.NumberofActiveMicrophones = '5'`
- Device could become unresponsive if queries are sent out back to back. It is recommended to have a delay of 2 or more seconds in between queries.

Supported Class and Example

EthernetClass

<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 31416, Model='Dicentis DCNM-LSYS')</code>

Active Microphone Directory Commands

- **Active Microphone Seat ID Results**
 - This command will display all the active microphone's Seat ID.
- **Active Microphone Participant ID Results**
 - This command will display all the active microphone's Participant ID.
- **Active Microphone Name Results**
 - This command will display all the active microphone's Seat Name.
- **Active Microphone Navigation**
 - This command is used for navigating through the Active Microphone Directory results. Please note, that this command will navigate through the results of both Active Microphone ID Results and Active Microphone Name Results simultaneously.
- **Active Microphone Update**
 - This command is used to update the results of the Active Microphone Directories and Microphone State.
 - This command is needed whenever the device is powered 'On' from an 'Off' state.

Device Configuration

Verify or configure that the Bosch Dicentis Webserver is assigned to a Seat and that it has the rights that you require:

- Has Power Off to power on/off the system.
- Manage meeting to control the microphones

DICENTIS Meeting application



3/20/2020 10:56 AM

☐ Configure mode (select on device) ☐ Rename seats

Seat Assignment

Select	Name	Status	Devices	VIP	Discuss	Manage meeting	Priority	VIS type	Voting (0)	Identification (0)	Language selection (2)	Camera
☐	Seat 1	✓	WIN-EIPPGPNEKF-WebClient	☐	☐	☒	☐	None	☒	☒	☒	None
☐	Seat 2	✓	WIN-EIPPGPNEKF-MeetingApplication	☐	☐	☒	☐	None	☒	☒	☒	None
☐	Seat 3	✓	DCNM-DE-0c1200 LEFT	☐	☒	☐	☐	None	☒	☒	☒	None
☐	Seat 4	✓	DCNM-DE-0c1200 RIGHT	☐	☒	☐	☐	None	☒	☒	☒	None
☐	Seat 5	✓	DCNM-DSL-0c11c0 LEFT	☐	☒	☐	☐	None	☒	☒	☒	None
☐	Seat 6	✓	DCNM-DSL-0c11c0 RIGHT	☐	☒	☐	☐	None	☒	☒	☒	None

Unassign Devices from Seat Delete empty seats Create Empty Seat

Select	Name	Dual us (1)	Type	Status	Serial Number	Version	Power Off	Volume Control
☐	Video Switcher		Video Switch	Ok	VideoSwitcher	3.20.22282		
☐	DCNM-DE-0c1200 RIGHT	☒	Discussion touch screen	Ok	045215461924022003	3.20.22282		
☐	DCNM-DSL-0c11c0 RIGHT	☒	Discussion language selector	Ok	045215X62001022038	3.20.22282		
☐	DCNM-DE-0c1200 LEFT	☒	Discussion touch screen	Ok	045215461924022003	3.20.22282		
☐	DCNM-DSL-0c11c0 LEFT	☒	Discussion language selector	Ok	045215X62001022038	3.20.22282		
☐	WIN-EIPPGPNEKF-ExternalVideoSwitch		External Video Switch	Ok	WIN-EIPPGPNEKF-ExternalVideoSwitch	3.20.22282		
☐	WIN-EIPPGPNEKF-WebClient		Meeting Application	Ok	WIN-EIPPGPNEKF-WebClient	3.20.22282	☒	☐
☐	WIN-EIPPGPNEKF-MeetingApplication		Meeting Application	Ok	WIN-EIPPGPNEKF-MeetingApplication	3.20.22282	☐	☐

Bosch Conference API Demonstrator is needed for the correct Seat and Participants IDs.

<https://<server>:31416/demonstrator/ConferenceAPIDemonstrator.html>

Conference API Demonstrator

Login Image Management Synoptic Layout Store Synoptic seat management Synoptic Discussion List Voting Result Power Mode Device Illumination Control Interpretation

User name:

Password:

Login Logout

Status: Login Success

Permissions:

- canViewSynoptic
- canEditSynoptic
- canSelect
- canAddDevicesToSeats
- canViewInterpretation
- canManageMeeting
- canControlInterpretation
- canSwitchSystemPowerOn
- canSwitchSystemPowerOff
- canViewVoting
- canViewRequestItems
- canViewFirstRequestOnSeat

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command ActiveMicrophoneNavigation	Value 'Up'	Value 'Down'	Value 'Page Up'
Value 'Page Down'			
# ActiveMicrophoneNavigation example InterfaceName.Set('ActiveMicrophoneNavigation', 'Up')			
Command ActiveMicrophoneUpdate ¹	Value None		
# ActiveMicrophoneUpdate example InterfaceName.Set('ActiveMicrophoneUpdate', None)			
Command MicrophoneState	Value 'On'	Value 'Off'	
Qualifier Key 'Seat ID'	Qualifier Value 'String'		
# MicrophoneState example InterfaceName.Set('MicrophoneState', 'On', {'Seat ID': 'String'})			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			

¹This command may be needed when the device is powered 'On' from an 'Off' state.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus, ActiveMicrophoneNameResults, ActiveMicrophoneParticipantIDResults, and ActiveMicrophoneSeatIDResults do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command ActiveMicrophoneNameResults	Value 'String'		
Qualifier Key 'Position'	Qualifier Value '1' – '10'		
# ActiveMicrophoneNameResults example Value = InterfaceName.ReadStatus('ActiveMicrophoneNameResults', {'Position': '1'}) InterfaceName.SubscribeStatus('ActiveMicrophoneNameResults', None, FeedbackHandler)			
Command ActiveMicrophoneParticipantIDResults	Value 'String'		
Qualifier Key 'Position'	Qualifier Value '1' – '10'		
# ActiveMicrophoneParticipantIDResults example Value = InterfaceName.ReadStatus('ActiveMicrophoneParticipantIDResults', {'Position': '1'}) InterfaceName.SubscribeStatus('ActiveMicrophoneParticipantIDResults', None, FeedbackHandler)			
Command ActiveMicrophoneSeatIDResults	Value 'String'		
Qualifier Key 'Position'	Qualifier Value '1' – '10'		
# ActiveMicrophoneSeatIDResults example Value = InterfaceName.ReadStatus('ActiveMicrophoneSeatIDResults', {'Position': '1'}) InterfaceName.SubscribeStatus('ActiveMicrophoneSeatIDResults', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command MicrophoneState	Value 'On'	Value 'Off'	Value 'Waiting'
Qualifier Key 'Seat ID'	Qualifier Value 'String'		
# MicrophoneState example InterfaceName.Update('MicrophoneState', {'Seat ID': 'String'})			

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Value = InterfaceName.ReadStatus('MicrophoneState', {'Seat ID': 'String'}) InterfaceName.SubscribeStatus('MicrophoneState', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command SeatInformation ¹	Value 'String'		
Qualifier Key 'Seat ID'	Qualifier Value 'String'		
Qualifier Key 'Type'	Qualifier Value 'Seat Name'	Qualifier Value 'Screen Line'	Qualifier Value 'Participant ID'
# SeatInformation example InterfaceName.Update('SeatInformation') InterfaceName.Update('SeatInformation', {'Seat ID': 'String', 'Type': 'Seat Name'}) Value = InterfaceName.ReadStatus('SeatInformation', {'Seat ID': 'String', 'Type': 'Seat Name'}) InterfaceName.SubscribeStatus('SeatInformation', None, FeedbackHandler)			

¹ Qualifiers are not required for the update.

Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 31416

Logon Credentials Supported: Yes

Multi-Connection Capabilities: Yes

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Power Off	<pre>{"parameters": {"powerMode": "poweredOff"}, "messageID": 0, "operation": "setsystempowermode"}</pre>
Power On	<pre>{"parameters": {"powerMode": "poweredOn"}, "messageID": 0, "operation": "setsystempowermode"}</pre>

Appendix B. Update Commands

Microphone State Seat ID testString	<pre>{"parameters": {}, "messageID": 0, "operation": "getDiscussionList"}</pre>
Power	<pre>{"parameters": {}, "messageID": 0, "operation": "getsystempowermode"}</pre>
Seat Information Type Screen Line Seat ID testString	<pre>{"parameters": {}, "messageID": 0, "operation": "getSeats"}</pre>
Seat Information Type Seat Name Seat ID testString	<pre>{"parameters": {}, "messageID": 0, "operation": "getSeats"}</pre>